INCH-POUND

MIL-M-14/5
2 April 1990

MILITARY SPECIFICATION SHEET

MOLDING COMPOUNDS, DIALLYL ISO-PHTHALATE THERMOSETTING, CONTAINING MINERAL FILLERS, GLASS OR POLYMERIC FILLERS

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-M-14.

Diallyl iso-phthalate resin molding compounds, having higher heat resistance than the corresponding ortho-phtalate compounds, shall be of the following types, as specified:

Type GII-30	–	Long glass fiber filler, nominal-impact strength, 3.0 foot-pounds per inch notch
Type GII-30F	–	Long glass fiber filler, flame-resistant, heat resistant, nominal-impact strength, 3.0 foot-pounds per inch notch
Type MIG	–	Mineral filler, heat-resistant
Type MIG-F	–	Mineral filler, flame-resistant, and heat-resistant
Type SIG	–	Short glass fiber filler, heat-resistant
Type SIG-F	–	Short glass fiber filler, flame-resistant, heat-resistant.

REQUIREMENTS

Qualification test requirements are specified in table I. Batch acceptant test requirements are specified in table II.

AMSC N/A FSC 9330

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NOTES:

Type GII-30. This type is a glass fiber-filled diallyl iso-phthalate resin molding compound of low-loss, high-dielectric strength, low shrinkage, excellent moisture resistance, and relatively high-impact strength.

Type GII-30F. This type is the same as GII-30 but is also flame-resistant.

Type MIG. This type is a mineral-filled diallyl iso-phthalate compound for use where good dielectric and heat resistance properties are required.

Type MIG-F. This type is a mineral-filled diallyl iso-phthalate compound for use where good dielectric and heat resistance properties in addition to flame resistance are required.

Type SIG. This type is a glass-filled diallyl iso-phthalate resin compound of low-loss, high dielectric strength, low shrinkage, heat resistance, good moisture resistance, and relatively low impact strength.

Type SIG-F. This type is a glass-filled diallyl iso-phthalate resin compound of low-loss, high dielectric strength, low shrinkage, flame-resistant, heat resistance, good moisture resistance, and relatively low-impact strength.

TABLE I. *Qualification test requirements for diallyl iso-phthalate resin molding compounds, mineral/glass filled.*

Requirement	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
			Mechanical/physical			
Compressive strength, endwise	24,000	24,000	20,000	20,000	24,000	24,000
Dimensional stability	0.1	0.1	0.2	0.2	0.2	0.2
Flexural strength	13,000	13,000	10,000	10,000	13,000	13,000
Heat deflection temperature	230	230	200	200	230	230
Heat resistance	170	170	130	130	140	140
Impact strength, side[1]	3.0	3.0	0.35	0.35	0.50	0.50
Tensile strength	6,000	6,000	6,000	6,000	6,000	6,000
Water absorption	0.5	0.5	0.7	0.7	0.5	0.5

See footnote at end of table.

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TABLE I. *Qualification test requirements for diallyl iso-phthalate resin molding compounds, mineral/glass filled – Continued.*

Requirement	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
Electrical						
Arc resistance	115	115	115	115	115	115
Dielectric breakdown:						
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	45	45	45	45	45	45
Short-time test (To be recorded as basis for determining initial voltage in step-by-step test)						
Step-by-step test	40	40	45	45	40	40
Dielectric constant:						
at 1 kilohertz	4.6	4.6	5.0	5.0	4.6	4.6
	4.7	4.7	5.0	5.0	4.7	4.7
at 1 megahertz	4.6	4.6	5.0	5.0	4.6	4.6
	4.7	4.7	5.0	5.0	4.7	4.7
Dielectric strength:						
Short-time test	325	325	325	325	325	325
Step-by-step test	300	300	300	300	300	300
Short-time test	325	325	325	325	325	325
Step-by-step test	300	300	300	300	300	300
Dissipation factor:						
at 1 kilohertz	0.010	0.010	0.019	0.019	0.010	0.010
	0.013	0.013	0.020	0.020	0.013	0.013
at 1 megahertz	0.018	0.018	0.019	0.019	0.015	0.015
	0.019	0.019	0.020	0.020	0.017	0.017
Surface resistance	5,000	5,000	5.0	5.0	10	5,000
Volume resistance	5,000	5,000	2.0	2.0	10	5,000
Water extract conductors	50×10^{-6}	50×10^{-6}	–	–	–	–
Combustion						
Flame resistance:						
Ignition time	–	90	–	90	–	90
Burning time	–	90	–	90	–	90
Flammability	–	V-0	–	V-0	–	V-0
Toxicity when heated:						
Carbon dioxide		15,000		15,000		
Carbon monoxide		1,000		1,000		
Ammonia		2,500		2,500		
Aldehydes as HCHO		50		50		
Cyanide as HCN		60		60		
Oxide of nitrogen at NO_2		100		100		
Hydrogen chloride		100		100		

[1]The side of the test specimen is that area formed by the chase of the mold.

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TABLE II. *Batch acceptance test requirements for diallyl iso-phthalate molding compounds, mineral/glass filled.*

Property to be tested	Type GII-30	Type GII-30F	Type MIG	Type MIG-F	Type SIG	Type SIG-F
	Performance requirement for each compound.					
Arc resistance	115	115	115	115	115	115
Dielectric constant at 1 megahertz	4.7	4.7	5.0	5.0	4.7	4.7
Dielectric strength, step-by-step	300	300	300	300	300	300
Dissipation factor at 1 megahertz	0.019	0.019	0.020	0.020	.017	0.017
Flexural strength	13,000	13,000	10,000	10,000	13,000	13,000
Impact strength, side[1]	3.0	3.0	0.35	0.35	0.50	0.50
Water absorption	0.5	0.5	0.7	0.7	0.5	0.5
Water extract conductors	50×10^{-6}	50×10^{-6}	–	–	–	–

[1]The side of the test specimen is that area formed by the chase of the mold.

Custodians:
Army – ME
Navy – SH
Air Force – 20

Review activities:
Army – ER, MI, MR, AR, EA
Navy – AS, OS
DLA – GS

User activities:
Navy – YD, EC

Preparing activity:
Navy - SH
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